Solar Irradiance Prediction: An Artificial Neural Network Approach versus Modified Pro-Energy

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***Abstract*—The accurate solar irradiance prediction can greatly improve the lifetime of energy harvesting wireless sensor networks (EH-WSNs) since they have limited battery capacity. In this paper, we show the prediction of solar irradiance with a feed-forward Artificial Neural Network (ANN) with the physics-informed features of the current and past irradiances that are cyclically encoded with the seasonal periodicity as well as the one-hour autoregressive lag. The model is trained and validated with a full year of irradiance measurements with a minute-resolution data from NREL (January 2019 to January 2020, 266,267 total raw samples, 4,419 hourly reading). The test results are R^2 of 0.9867, RMSE of 20.027 W/m². The baseline method is the standard Pro-Energy method and its modified for additional weight coefficient of the moving average, which referred as Modified Pro-Energy. The original Pro-Energy and its modified version have a lower R^2 and higher RMSE than the ANN one, where the Modified Pro-Energy has R^2 0.8203, RMSE 73.7216 W/m² (nearly about 4 times larger). The ANN model shows strong performance also on the cloud-interrupted days, while the baseline methods become unreliable during that period. The results show that the metrics of predicting solar irradiance favor the ANN approach over the existing ones.**

***Index Terms*—Solar Irradiance Prediction, Artificial Neural Network, Modified Pro-Energy, Energy Harvesting, Wireless Sensor Networks, IoT, Deep Learning, Feature Engineering.**

I. Introduction

Wireless Sensor Networks (WSNs) are widely used in areas, such as tracking the environment , smart cities and other IoT application.[12], [13]. These networks are typically composed of dozens to hundreds of sensor nodes scattered across a region, each equipped with sensing, processing, and radio communication hardware. The small embedded batteries powering each node are eventually depleted, and physically replacing them in remote locations is expensive and not easy. [1], [10].

By equipping each sensor node with a small photovoltaic (PV) panel and a rechargeable battery , networks can in theory operate very long [3]. However, solar energy is unpredictable, as solar irradiance can change hourly or daily or seasonally based on several environmental factors. If the network's power manager does not know in advance how much energy is coming, it cannot make good decisions about transmission power, duty cycling, or data aggregation [14]

So a reliable forecast of solar irradiance is needed for sensor , as it helps in making good decisions about power usage and also there is need of light weight algorithms , as computing power is less on these sensors.

A. Motivation

Energy Harvesting (EH) helps in improving the limited energy(low capacity batteries), in WSN by making renewable sources like solar and wind into electricity. It is cost-effective and provides a potentially unlimited energy supply. An EH system typically includes components like rechargeable batteries, supercapacitors, transducers, energy predictors, and power management units. Since energy sources such as solar and wind depend on changing weather conditions, the energy supply is unpredictable. That's why energy prediction is important to manage and optimise power usage in sensor nodes efficiently. Also, our main concern is to make hourly solar irradiance prediction, which is highly variable and usable in terms of battery power management compared to shorter time slots(1,2 or 10 minutes)

B. Related Works

A number of approaches have been proposed for solar energy prediction in wireless sensor networks.

Statistical Moving Average Methods. The methodological approach adopted for the problem of early weather/energy prediction, when faced with constrained devices, can be divided into two overall categories. The first category is of course general purpose algorithms researched and engineered by computer scientists and math faculties. These algorithms have been fine tuned to work perfectly on high performance servers with lots of CPUs and lots of memory. Although, these could be analyzed in terms of efficiency to understand their efficiency behavior when scaling down to fit on low performance computational devices with lots of constraints with respect to computational capacity and memory size. For example, let's take a look at (Exponential) Weighted Moving Average (EWMA). According to the studies of Kosunalp [6], [7], this primitive approach is computed by blending yesterday's actual measurement with a running historical average using a scalar weight α. Thus, the computation of the algorithm is kept to the bare minimum. This is absolutely fine and helpful when the weather is stable, but it has no mechanism for sharp weather transitions. This failure can also be found in the the next algorithm that we will review which is a simple extension of EWMA, called WCMA. The main difference, as can be expected by it's name, is that WCMA introduces a new-dimensional dimensionless gap coefficient GAPk. GAPk compares the current day's energy rate against a historical mean. And since it so easy to compute GAPk

that it grows linearly with the lookback window, it proves to be less suitable for resource-constrained nodes [1]. As a closing note for this category, we would also like to mention a novel approach by Cammarano et al. [5], called Pro-Energy. In order to predict whether the energy output of the current day turns out to be reasonable, Cammarano et al[5]. store several preset canonical daily profiles in a library. These profiles are typically, sunny, cloudy and mixed, in order to forecast the output rates under the above conditions. The current day's observations are then compared against these profiles. The energy profile with minimum MAE against the current daily observations until a lookahead point are considered as the best guess and returned as the predicted energy output for the remaining part of the day. Although the approach performs well on typical days, it turns into being extremely weak when today's weather has no close analogue in the library.

Profile-Based Methods, Modified Pro-Energy.

To further explore these findings, comparing the Pro-Energy approach to a modified Pro-Energy approach with a similar structure and purpose, based on suggestions by Sah et al[1]. to introduce a term ρ(d, n) to perform live error-corrections and modify subsequent predictions based on accumulated predictions errors made so far [1]. When predicting the power output for a given day d on the nth hour ($0 <= n < 24$) based on its seasonal profile, the monthly statistics (mean/standard deviation) of the past at the same hour are used to estimate this prediction error. Using its calculation, the next prediction made at a future hour will be updated by the calculated error of the past hour, alleviating the disadvantage of the standard Pro-Energy approach. This modified Pro-Energy approach can be expected to outperform the EWMA and WCMA approaches, but these improvements may come to the expense of using profiles to predict output all together.

To this end, the second approach by Cammarano et al.[5] is revisited. The original approach is based on the Pro-Energy approach to use a library of canonical daily profiles to match the typical forms for energy production that can be expected for different weather situations such as sunny, cloudy, mixed etc. By storing various combinations of daily weather profiles, the library is reduced to the minimum required number of canonical profiles, while the final generation of the profile can vary from day to day based on which canonical profiles are selected, enabling to accurately capture the structure of typical days. The benefit of this approach lies in being able to accurately perform predictions for days that look typical with regards to prior weather conditions, but unfortunately suffer from a major disadvantage in case the weather does not turn out to be typical. Today's weather may have no close analogue in the stored library, in case the weather being observed today was not encountered during the time the library of canonical profiles was built. In these instances, this approach will yield a poor match producing an accurate profile and obviously the disadvantage of the not being able to utilize the heuristic advantages of machinery learning based on previously collected time series identification data.

The first proposed method coupled with time series identification of day ahead energy output was not optimal. Depending on the ever-changing weather, there were always situations in which discrepancies in estimated by successive adjustments prolonged erroneous predictions either due to the inability to output a profile with the anticipated structure of today or the recently developed weather related malfunctioning with the profile-predicting approach. A rather promising approach that was able to output the right prediction in case the simulated weather was different than the normal, but was much more reliant on the availability of an accurate description of the weather led to discrepancies that could have been solves simply by remaining with the time series predictive approach and retaining the option for making small adjustments without sacrificing the advantages of machinery learning.

Machine Learning Approach.

The field of solar-related prediction and control has investigated both reinforcement learning and deep learning methods. Hsu et al. [4] tested RL agents that learn duty-cycle control policies without an explicit prediction model. Kosunalp [6] utilized a Q-learning predictor in which the harvesting rate is casted as a Markov decision process. There are certain limitations of the reinforcement learning approach, mainly that it can take considerable time for convergence, and the learned models are limited in their ability to generalize over seasons and geographic locations. In contrast, neural networks can capture complex, non-stationary irradiance dynamics to a desirable accuracy, and they have proven to be very successful for the task of solar forecasting

There are some research done related for EH-WSN , like Ashraf et al. [2] has given a prediction method where it works on short-term and long-term data , named ASIM and also it uses nonlinear model to show throughput in network. Yang et al. [3] came with solar-powered WSN built for self-protection and self-healing ,making it handle prediction errors with minimum risk of WSN failure. Hsu et al. [4] have demonstrated Reinforcement learning identifies better power management in networks through various states. Kosunalp [6] proposed Q-learning-based solar energy prediction. Dehwah et al. [8] used two algorithms , mixed integer linear programming(MILP) and dynamic programming., Ganjewar et al. [20] with his transmission technique using hierarchical least-mean square (HLMS), can predict sensed data in the WSN which shown effective neutral power management while Peng et al. [21] focused on energy-neutral power management. One focus of the proposed algorithm is to manage the energy budget of the sensor node. Gorlatova et al. [22] came up with device to control uncontrollable energy resources, it is based on the long-term indoor solar radiant energy level , primary considered for battery energy. Herrería-Alonso et al. [23] observed that different factors such as angles of sun light also affects energy prediction. Zou et al. [9] have proposed a cluster-based routing technique using prediction technique optimization in EH-WSNs. Overall,researchers demonstrate that accurate prediction and intelligent energy management are critical to preventing node failure and ensuring efficient, sustainable operation of EH-WSNs. Leading to base problem.

C. Problem Statement

Let $H(d, n)$, measured in W/m^2, denote the harvested energy at day d and time slot n. The prediction task is to estimate

$\hat{E}(d,n)$(the expected available energy for the *next* slot) using only information available up to the present moment. As next hour data is useful for cells in WSN , to make decisions like to either save or spend remaining energy. A good predictor has to simultaneously satisfy two competing objectives: it must be accurate enough to prevent energy outages (under-predictions drain the battery unexpectedly), yet conservative enough to avoid wasteful over-transmission (over-predictions lead the node to spend energy that never actually arrives).

More precisely, we want to minimise the slot-averaged prediction error:

$$\bar{\varepsilon} = \frac{1}{D \cdot N} \sum_{d=1}^{D} \sum_{n=1}^{N} \left| \hat{E}(d,n) - H(d,n) \right| \tag{1}$$

across all D test days, each divided into N time slots.

These factors make the problem difficult.First, weather transitions (such as sudden cloud cover in particular) introduce sharp, non-stationary drops in irradiance that simple averaging methods simply cannot track. Second, sensor nodes are typically deployed without access to external weather forecasts, as the periodic transmission of cloud-based data is energy expensive for battery-constrained devices. Furthermore, local sensors often possess superior granular knowledge of the specific micro-climate than coarse global models, meaning the predictor must operate autonomously using only on-device historical measurements.

D. Contribution

We will show that well-designed ANNs can learn the non-stationary and multi-scale patterns without relying on template matching (like other earlier methods), as a result removing the historical-lock problem altogether.The proposed framework operates on a context-optimized feature extraction pipeline. As a result, the following three inputs are used to improve prediction accuracy:

- The normalized irradiance signals.
- A set of cyclic temporal encodings that improve the ANN's understanding of the cyclic component of the signal.
- The autoregressive lag of the irradiance signal to capture temporal dependencies.

By making 11 channels down to 4 channels using correlation table so that no two highly correlated(> 0.98) channels fed into the ANN reducing redundancy plus 3 feature engineering channels which totals 7 features, we enable the model to focus on the most relevant correlation for the application at hand. The proposed solution is a lightweight compact two-hidden-layer feed-forward ANN trained on 4 features obtained from the 4,419-sample hourly dataset (see Section 3) and 3 engineered features . The I/O of this model can be simplified as follows:

- The neural network comprises two hidden layers:
- **Linear function** : defined by node weights w and bias p.
- **Non-linear activation** : such as Sigmoid or Rectified Linear Unit (ReLU).

For reference to the mathematical formulation for the Modified Pro-Energy baseline implemented in our work we provide the full mathematical derivation to obtain the Modified Pro-Energy Indicator formula (see Section 3) that serves as the proper baseline in this application context. In addition, to fully behave according to the described weighted profile selection and error correction dynamics we provide the pseudocode with complexity analysis of this algorithm. To evaluate the accuracy of the proposed architecture we have achieved recurring success in analyzing real NREL irradiance. As a result, we have shown that:

- The quality of the predictions generated can be improved by focusing on nearby spatial patterns to predict irregular structures of light that are likely not to repeat but share combined gradients over larger areas in space.
- Works better then existing models in high variance days. (variance(hourly datapoint) ¿ variance(minute level) - still for hourly ANN perform better)

Section II, Preliminaries and has the mathematical model of solar harvesting system and other methods, EWMA, Modified Pro-Energy. Section III ,The proposed algorithm ANN. Section IV, the experimental results of ANN with comparison of other methods. Section V states the presence of certain limitations. Practically, the simulated results shows applicability of neural networks in solar harvesting,minor limitations and simulations will be included to prove the concept of neural networks at large scale. Section VI, Conclusion.

II. Preliminaries and Algorithm Evolution

A. Solar Harvesting System Model

Each sensor node in the target network is powered by a rechargeable battery coupled to a photovoltaic harvesting module. A day is divided into N equal-length time slots of duration T, where $N \cdot T = 24$ hours. The harvested energy at slot n of day d is $H(d,n)$ in W/m^2, and the battery energy level after that slot is $E_L(d,n)$.

The PV cell output current follows the standard diode equation, which captures the nonlinear dependence on irradiance, temperature, and load [11]:

$$I_{\text{solar}} = I_{\text{light}} - I_0 \Big(e^{\frac{V_{\text{solar}} + I_{\text{solar}} R_S}{a}} - 1 \Big) - \frac{V_{\text{solar}} + I_{\text{solar}} R_S}{R_{SH}} \tag{2}$$

where I_{light} is the photogenerated current (proportional to irradiance), I_0 is the reverse saturation current, R_S and R_{SH} are the series and shunt resistances, and a is the modified ideal factor:

$$a = \frac{N_S \gamma k T_C}{q} \tag{3}$$

Here N_S is the number of series cells, γ is the ideality factor, k is Boltzmann's constant, T_C is cell temperature, and q is the electron charge. This formulation motivates using irradiance (W/m^2) as the primary prediction target, since it is the fundamental physical quantity that drives all the nonlinear terms in the equation above.

The energy predictor sits between the PV module and the PMU. Its estimate $\hat{E}(d,n)$ gives the PMU time to adjust transmission power and sleep scheduling before the next slot

arrives. The implication of EH theory is fairly direct: when prediction error is small, the battery energy level E_L stays elevated, network throughput increases, and overall lifetime extends [1]. Even a systematically biased predictor, on the other hand, can cause the network to repeatedly deplete or under-utilise its battery.

Remark 1: The harvesting rate is dynamic and varies with seasonal conditions. At solar noon on a clear summer day, the harvesting rate can reach $D_{\max}(h_r^+)$; during nighttime or heavy overcast, it falls to $D_{\max}(h_r^-)$. A practical predictor must handle this entire range without special-casing either extreme.

B. Historical Baseline Evolution

1) EWMA. The Exponential Weighted Moving Average estimates energy at slot n on day d as:

$$E(d,n) = \alpha E(d-1,n) + (1-\alpha)H(d-1,n) \tag{4}$$

The scalar $\alpha \in (0,1)$ controls how quickly older information is discounted. EWMA is simple and inexpensive, but it carries memory of only a single past day and has no weather-state model, so it cannot react to sudden changes.

2) WCMA. The Weather-Conditioned Moving Average stores a $D \times N$ matrix of past energy readings and introduces a dimensionless weather ratio GAP_k:

$$E(d,n) = \alpha H + (1-\alpha)M_D(d,n) \cdot GAP_k \tag{5}$$

where $M_D(d,n) = \frac{1}{D}\sum_{i=d-D}^{d-1} E(i,n)$ is the D-day mean for slot n. The ratio $V_k = E(d,n-K+k)/M_D(d,n-K+k)$ measures how today's recent harvesting rate compares to the historical mean; these ratios are then combined with recency weights $F_k = k/K$ to form:

$$GAP_k = \frac{\mathbf{V}\cdot\mathbf{F}}{\sum F} = \frac{\sum_{k=1}^{K} v_k f_k}{\sum_{k=1}^{K} f_k} \tag{6}$$

WCMA adapts faster to weather transitions than EWMA, but computing GAP_k grows with both D and K.

3) Pro-Energy. Profile Energy maintains a library of canonical sunny, cloudy, and mixed daily curves. At each slot, it retrieves the closest profile by MAE against the current day's data observed so far:

$$MAE(E_d, C) = \frac{1}{K}\sum_{i=t-K+1}^{t} |C_i - E_{d,i}| \tag{7}$$

The profile with the smallest MAE is used for prediction. Pro-Energy outperforms both moving-average methods on typical days, but when the current day has no close analogue in the library, the method is left with a poor match and correspondingly large errors.

C. The Leap to Deep Learning

All three methods described above are essentially *historical lookup* approachesthey can only anticipate energy patterns that have appeared in some form before. A neural network operates on an entirely different principle. Rather than memorising past days, it learns the *mapping function* from inputs (time of day, season, recent irradiance) to outputs (next-hour irradiance). Once that mapping is learned, the model can handle any combination of inputs including weather conditions that never appeared during training. This fundamental generalisation capability is the core motivation for our approach.

III. Proposed Framework and Architecture

A. Data Engine and Preprocessing Pipeline

1) Data Overview: Our dataset was obtained from the National Renewable Energy Laboratory (NREL) Measurement and Instrumentation Data Center [19]. The raw data comprises 266,267 solar irradiance readings collected at one-minute intervals from January 2019 through January 2020, spanning a complete annual cycle of seasonal and diurnal variation. Several sensor modalities were recorded simultaneously at the same site, including pyranometers of different grades (CMP22, CMP11, PSP), a UV-B radiometer, and sensors oriented at various angles. This multi-sensor setup provides rich redundancy, but also introduces strong inter-feature correlation that needs to be handled carefully before training.

Figure 1 shows the distribution of hourly-averaged GHI values across the full dataset. The distribution is notably bimodal: a large concentration near zero corresponds to nighttime and heavily overcast hours, while a secondary cluster around 400–700 W/m^2 captures productive daytime hours on clear days. Near solar noon on bright summer days, peak irradiance can exceed 1,000 W/m^2. This dynamic rangespanning more than three orders of magnitudeis one reason simple linear averaging methods struggle.

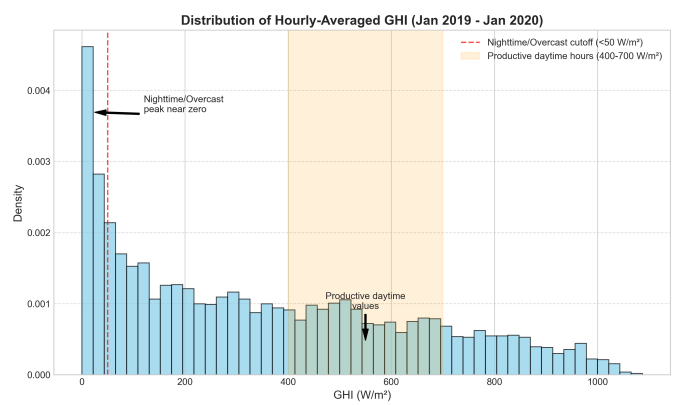

Fig. 1: Distribution of hourly-averaged GHI values across the full January 2019 – January 2020 dataset. The bimodal structure reflects nighttime/overcast hours near zero and productive daytime hours in the 400–700 W/m^2 range.

Table I: Actual Dataset overview with vertical and horizontal borders.

Datetime	CMP22	UVB-1	90-West	CUVB2
2019-01-26 07:00:00	24.05	0.00	15.74	0.00
2019-01-26 08:00:00	125.37	0.05	76.24	0.02
2019-01-26 09:00:00	381.46	0.18	199.26	0.20
2019-01-26 10:00:00	430.22	0.31	217.49	0.15
2019-01-26 11:00:00	541.48	0.45	211.03	0.36
2019-01-26 12:00:00	582.79	0.49	261.51	0.46
2019-01-26 13:00:00	534.78	0.40	454.48	0.39
2019-01-26 14:00:00	422.07	0.25	595.61	0.26
2019-01-26 15:00:00	281.61	0.10	609.38	0.12
2019-01-26 16:00:00	80.62	0.02	204.49	0.00
2019-01-26 17:00:00	11.39	0.00	9.75	0.00
2019-01-27 07:00:00	67.72	0.01	40.77	0.00
2019-01-27 08:00:00	177.22	0.05	82.93	0.04
2019-01-27 09:00:00	289.57	0.18	123.35	0.11
2019-01-27 10:00:00	244.89	0.24	115.36	0.01

Table II: Dataset Summary Statistics

Property	Value
Measurement interval (raw)	1 minute
Date range	Jan 2019 – Jan 2020
Raw records	266,267
Aggregated hourly records	4,419
Raw sensor channels	11
Channels after pruning	4 physical + 3 temporal
Target variable	Global CMP22 (GHI, W/m^2)
GHI range	0 – ≈1,100 W/m^2

2) Preprocessing Steps: The first step was temporal aggregation. One-minute readings were averaged into hourly windows, which simultaneously removes high-frequency sensor noise and aligns the prediction granularity with the time-slot model used by WSN energy managers. Nighttime hours structurally zero irradiance were kept in the dataset because they provide important context for the lag feature. After discarding rows containing missing or corrupted sensor values, we were left with 4,419 valid hourly records.

The second step was correlation-based feature pruning. The NREL dataset includes multiple channels measuring physically identical quantities from instruments of different grades (for instance, Global CMP22, Global CMP22-1, and Global CMP11 have Pearson correlations exceeding 0.99). Retaining all of them adds no useful information while inflating input dimensionality unnecessarily, and that in turn can destabilise gradient-based training. We computed the full Pearson correlation matrix across all 11 channels (Figure 2) and removed any channel with a correlation above 0.98 with another retained channel. This reduced the physical input set from 11 down to 4 distinct measurements.

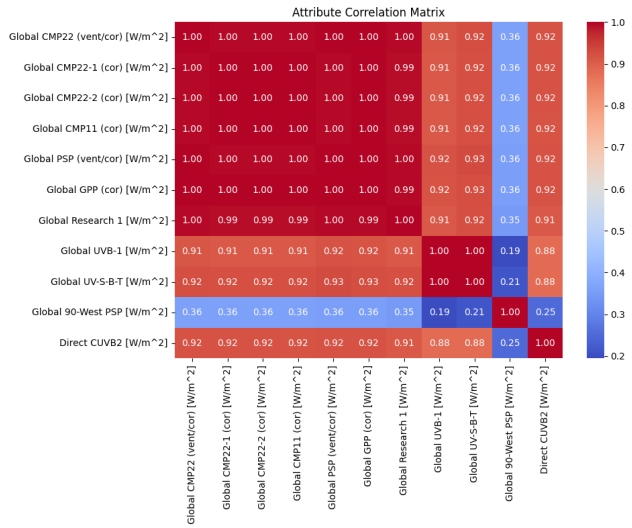

Fig. 2: Pearson correlation heatmap across all 11 raw sensor channels before pruning. Deep-red channel pairs (correlation > 0.98) were removed to eliminate redundancy and improve training stability.

The third step was standard score normalisation. All retained numerical features were scaled to zero mean and unit variance using statistics computed exclusively on the training fold, so no information from the validation or test sets leaks into preprocessing.

3) Cyclic Temporal Encoding: A straightforward integer representation of time (hour = 0, 1, . . . , 23) creates an artificial discontinuity: the model would treat hour 23 and hour 0 as maximally distant when they are actually adjacent. The same issue arises at the December–January year boundary. We address this with sine/cosine projections that restore cyclic continuity:

$$\sin_{\text{hour}} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad \cos_{\text{hour}} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right) \tag{8}$$

$$\begin{aligned} \sin_{\text{doy}} &= \sin\left(\frac{2\pi \cdot \text{dayOfYear}}{365}\right) \\ \cos_{\text{doy}} &= \cos\left(\frac{2\pi \cdot \text{dayOfYear}}{365}\right) \end{aligned} \tag{9}$$

Each pair maps every time step onto a point on the unit circle, ensuring that adjacent hours and adjacent days of the year are always geometrically close in the input space [18].

4) Autoregressive Lag Feature: To give the model a sense of short-term momentum, we include a one-hour lag of the target:

$$\text{GHI_lag}_1(t) = \text{GHI}(t-1) \tag{10}$$

Irradiance at hour t is typically strongly correlated with irradiance at hour $t-1$, since atmospheric conditions do not change instantaneously. This single feature carries substantial predictive value and effectively gives the model a running start before it tries to extrapolate into the next slot.

B. Feature Engineering Summary

After all preprocessing, the final input feature vector $\mathbf{x} \in \mathbb{R}^7$ contains: four physically distinct sensor measurements (Global CMP22, Global UVB-1, Global 90-West PSP, Direct CUVB2), two diurnal cyclic encodings ($\sin_{\text{hour}}$, $\cos_{\text{hour}}$), two seasonal cyclic encodings ($\sin_{\text{doy}}$, $\cos_{\text{doy}}$), and the one-hour autoregressive lag. Table III summarises each feature and its purpose.

Table III: Selected Attributes and Engineered Features for ANN Training

Attribute / Feature	Description
Global CMP22 (vent/cor)	Primary target: GHI in W/m^2. Most precise pyranometer channel.
Global UVB-1	UV-B radiation (W/m^2). Orthogonal spectral information retained after pruning.
Global 90-West PSP	West-facing irradiance. Captures afternoon sun-angle geometry.
Direct CUVB2	Direct UV component. Differs from diffuse GHI under cloud cover.
$\sin_{\text{hour}}$, $\cos_{\text{hour}}$	Diurnal cyclic encoding. Eliminates 23/0 hour discontinuity.
$\sin_{\text{doy}}$, $\cos_{\text{doy}}$	Seasonal cyclic encoding across the full year.
GHI_lag_1	GHI from the previous hour. Provides autoregressive momentum.

C. Artificial Neural Network Architecture

1) Design Rationale: With only 4,419 training samples, the dominant architectural risk is overfitting. A deep or wide network would likely memorise the training data rather than learning patterns that generalise to unseen hours. For this reason, we deliberately chose a compact design: two hidden layers with 64 and 32 units respectively. This gives the network enough capacity to capture nonlinear feature interactions while keeping the parameter count small enough that early stopping alone provides adequate regularisation.

2) Layer Specification: The network processes data through the following sequence:

1) **Input Layer:** Receives $\mathbf{x} \in \mathbb{R}^7$, pre-normalised to zero mean and unit variance on the training set.
2) **Hidden Layer 1:** 64 neurons with ReLU activations. The pre-activation is $\mathbf{z}^{(1)} = \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}$, and the output is $\mathbf{h}^{(1)} = \max(0, \mathbf{z}^{(1)})$.
3) **Hidden Layer 2:** 32 ReLU neurons, giving $\mathbf{h}^{(2)} = \max(0, \mathbf{W}^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)})$.
4) **Output Layer:** A single linear neuron producing the scalar prediction $\hat{y} = \mathbf{w}^{(3)\top}\mathbf{h}^{(2)} + b^{(3)}$. Linear activation is appropriate here because irradiance is an unbounded continuous variable.

The total number of trainable parameters works out to $(7 \times 64 + 64) + (64 \times 32 + 32) + (32 + 1) = 2{,}657$, a figure well matched to the dataset size.

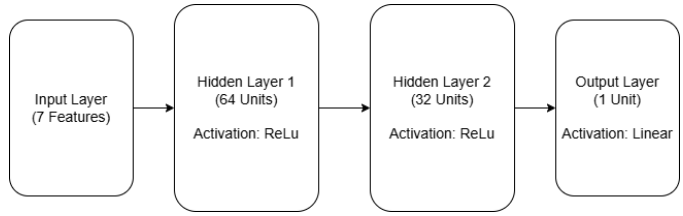

Fig. 3: Block diagram of the proposed ANN. Seven engineered features feed into two hidden layers (64 and 32 ReLU units) and a single linear output neuron predicting GHI in W/m^2.

3) Loss Function and Optimisation: The network minimises Mean Squared Error over each mini-batch of $N_b = 32$ samples:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N_b} \sum_{i=1}^{N_b} (y_i - \hat{y}_i)^2 \tag{11}$$

We use the Adam optimiser [15] with its default hyperparameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$), which adapts per-parameter learning rates from first and second moment gradient estimates and converges substantially faster than vanilla SGD on moderate-sized regression problems.

Training is allowed to run for up to 50 epochs. An early-stopping callback watches validation MSE and halts training if no improvement appears for 5 consecutive epochs, restoring the best-performing weights found up to that point. This prevents overfitting without requiring the practitioner to manually tune training duration.

D. Modified Pro-Energy: Full Formulation

1) Profile Classification: The Modified Pro-Energy baseline [1] maintains a $D \times N$ energy matrix $\mathbf{E}$ covering the most recent D days, each divided into N slots. Each stored day is assigned to one of three weather archetypes based on a threshold $\theta = 0.4$:

$$E'_{\text{sunny}} = \sum_{i=1}^{D} \sum_{j=1}^{N} E_{\text{sunny}}(i,j) \tag{12}$$

$$E'_{\text{cloudy}} = \sum_{i=1}^{D} \sum_{j=1}^{N} E_{\text{cloudy}}(i,j) \tag{13}$$

$$E'_{\text{mix}} = \sum_{i=1}^{D} \sum_{j=1}^{N} E_{\text{mix}}(i,j) \tag{14}$$

A day is labelled sunny if $\sum_j E(i,j) > (1-\theta)\bar{E}$, cloudy if $\sum_j E(i,j) < (1+\theta)\bar{E}$, and mixed otherwise, where $\bar{E}$ is the long-term average daily energy.

2) Weighted Profile Selection: Rather than relying on only the single best-matching past day, Modified Pro-Energy blends P profiles weighted by similarity. The MAE between a stored day E_d and the current day's observations C over the last K slots is:

$$MAE(E_d, C) = \min_{E_d \in \mathbf{E}} \frac{1}{K} \sum_{i=t-K+1}^{t} |C_i - E_{d,i}| \tag{15}$$

The weight assigned to the j-th most similar profile is:

$$W_j = 1 - \frac{MAE(E_{d_j}, C)}{\sum_{j=1}^{P} MAE(E_{d_j}, C)} \tag{16}$$

and the combined weighted profile is:

$$WP = \frac{\sum_{j=0}^{P} W_j \cdot E_{d_j}}{P-1} \tag{17}$$

Days with lower MAE (higher similarity) naturally receive greater weight in the blend.

3) Error-Correcting Prediction: The prediction for slot n on day d is:

$$\hat{E}(d,n) = \alpha H(d, n-1) + (1-\alpha) WP - \rho(d,n) \tag{18}$$

where $H(d, n-1)$ is the measured energy from the previous slot. The correction term ρ tracks accumulated prediction error within the current day:

$$\rho(d,n) = \beta\rho(d,n-1) + (1-\beta)\left[\hat{E}(d,n-1) - E_{\text{actual}}(d,n-1)\right] \tag{19}$$

The parameter β controls how long the error memory lasts. When the algorithm over-predicts, ρ grows and pulls the next prediction downward; when it under-predicts, ρ turns negative and nudges the next prediction upward. Following [1], we use $\alpha = 0.5$, $\beta = 0.2$, and $D = 10$.

4) Algorithm and Complexity: Algorithm 1 lays out the complete procedure.

Lemma 1: The time complexity of the Modified Pro-Energy algorithm is $\mathcal{O}(D \times N)$ per day.

Three stages dominate the runtime: (1) weather classification scans all D stored days at $O(D)$ cost; (2) MAE computation to rank profiles costs $O(D)$ per slot (only K recent slots are checked), which is $O(D)$ overall since K¡¡N; (3) the per-slot prediction loop runs for N slots and performs $O(D)$ work per slot to compute WP and update ρ. The total is $O(D) + O(D) + O(D \times N) = O(D \times N)$.

Remark 2: For $D = 10$ and $N = 24$ hourly slots, this amounts to 240 multiply-accumulate operations per daywell within reach of a modern ARM Cortex-M4 microcontroller at negligible battery cost.

Algorithm 1: Modified Pro-Energy Prediction

Input: α, β, D, $\mathbf{E}(i,j)$, threshold θ
Output: Predicted energy sequence $\hat{E}[d,\cdot]$ for day d

```
Classify each stored day as sunny, cloudy, or mixed using θ;
Initialise: α ← 0.5, β ← 0.2, D ← 10, θ ← 0.4;
for i ← 0 to D do
    if Σ_j E(i,j) > (1 − θ)Ē then
        sunny day
    else
        if Σ_j E(i,j) < (1 + θ)Ē then
            cloudy day
        else
            mixed day
        end
    end
end
Compute MAE(E_d, C) for each of the D stored days;
Sort stored days by ascending MAE (descending similarity);
for j ← 0 to P do
    W_j ← 1 − MAE(E_{d_j}, C)/Σ_j MAE(E_{d_j}, C);
    WP ← Σ_j W_j · E_{d_j}/(P − 1);
end
Initialise error corrector ρ ← 0;
for each slot s ← 0 to N do
    ρ ← β·ρ + (1 − β)·[Ê(d, s − 1) − E_actual(d, s − 1)];
    δ ← E^T[slot][D − 1 : 0] · W;
    Ê[d, s] ← α · E(d, s − 1) + (1 − α) · δ − ρ;
end
```

E. Data Splitting Strategy

The two models require different evaluation protocols because of how differently they operate internally.

For the ANN, which treats each samples independently, we randomly shuffled all 4,419 hourly records and applied a 70% / 20% / 10% split into training, validation, and test sets. Random splitting is valid here since the ANN's prediction depends only on the input feature vector for that hour.

For Modified Pro-Energy, random shuffling is not an option-the algorithm explicitly requires sequential complete days to populate its profile library and run within-day error correction. We therefore evaluated it chronologically: the first 90% of calendar days served as the profile history, and predictions were made for the remaining 10% of days in order. Both models were evaluated over comparable calendar periods to keep the comparison fair.

IV. Results and Performance Analysis

A. Evaluation Metrics

We evaluate predictions using four different measures. RMSE and MAE tell you how far off the predictions are in real physical units (W/m²). MSE punishes large errors more heavily, so it's useful for catching really bad predictions. R²

tells you how well the model explains the overall variation in the data, on a scale that works regardless of the units involved.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{20}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{21}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{22}$$

B. Overall Performance Comparison

The ANN achieves an R^2 of 0.9867, which means it explains 98.67% of all variance in the irradiance of test set. Its RMSE of 20.027 W/m^2 amounts to roughly a 2% error relative to a typical clear-day peak near 1,000 W/m^2. The Modified Pro-Energy model, despite its error-correction mechanism, explains only 67.35% of the variance and produces an RMSE nearly five times larger.

Figure 4 visualizes this gap.

Fig. 4: Bar chart comparing RMSE, MAE, and R^2 between the proposed ANN and the Modified Pro-Energy. (Lower is better for RMSE and MAE) (higher is better for R^2).

C. Training Convergence

The ANN converged smoothly . The validation loss stayed close to the training loss the whole way through, which means the model was picking up genuine patterns rather than just memorizing the training data. It also stopped early ,around 35 epochs out of a possible 50 , which suggests the small, lean architecture was just the right size for the amount of data available. When bigger or wider networks were tried in earlier experiments, they clearly started overfitting, which confirmed that keeping the model compact was the right call

D. Time-Series Analysis

1) Full Test Period: Figure 5 shows the prediction trajectories of both models across the entire test period, plotted against ground-truth irradiance. The ANN's output forms a smooth curve that closely follows the true signal, correctly reproducing both daily peak shapes and quiet overnight periods. The Modified Pro-Energy trace shows characteristic profile-matching artefacts: it frequently locks onto the wrong historical template and produces staircase-like predictions that lag real irradiance transitions.

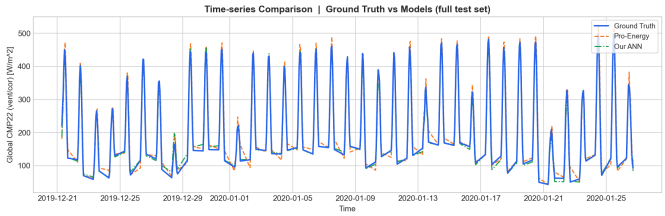

Fig. 5: Full test-period comparison of ground-truth GHI against ANN and Modified Pro-Energy predictions.

2) Clear Sky Day: On a stable, cloud day like December 28, 2019 (Figure 6), both models perform well. The irradiance follows a bell-shaped curve centred on solar noon, which closely resembles the sunny archetype in the Pro-Energy profile library. In this scenario, profile matching is correct by construction, and the error-correction term ρ has little work to do. The ANN also tracks the curve accurately, with better alignment near the peaks.

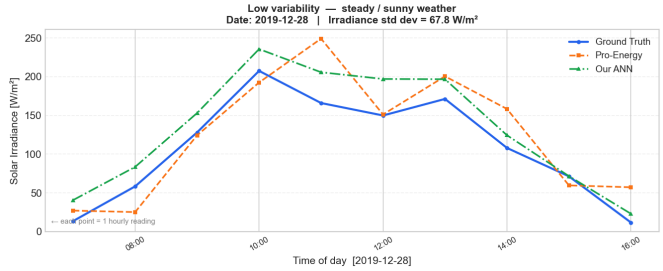

Fig. 6: Clea sky day (December 28, 2019): ANN model closely resembles to actual data then Modified pro energy model.

3) High-Variance Day: January 24, 2020 tells a very different story. Repeated cloud-break events caused irradiance to swing rapidly between near-zero and several hundred W/m^2 within single-hour windows. As Figure 7 shows, Modified Pro-Energy commits to a profile match early in the day and then struggles as the error-correction term tries but ultimately failsto compensate for an increasingly wrong profile choice. The result is a prediction that remains systematically offset from reality for much of the afternoon.

The ANN, having no profile to commit to, simply processes the current feature vector at each step. While it is not perfect on the sharpest transitions, its predictions track the volatile ground truth far more closely throughout the day, with no systematic offset and no profile-locking artefact in sight.

Table IV: Overall Quantitative Performance Metrics Comparison

Metric	ANN (Proposed)	Modified Pro-Energy	EWMA	WCMA	Standard Pro-Energy
RMSE (W/m^2)	**20.027**	73.721	124.273	96.087	152.262
MAE (W/m^2)	**19.910**	51.552	61.050	51.402	85.670
R^2	**0.9867**	0.820	0.7977	0.8793	0.6959

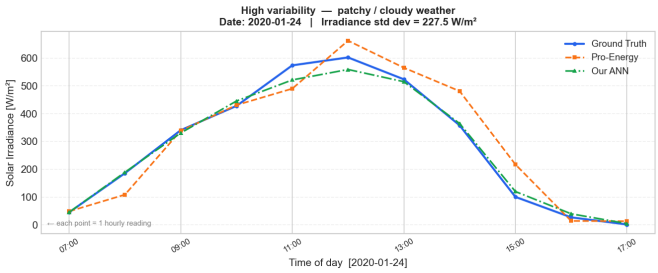

Fig. 7: High-variance day (January 24, 2020): Modified Pro-Energy locks onto the wrong profile and cannot adapt to rapid cloud events. The ANN follows each transition far more closely.

This asymmetry is arguably the most practically important finding of the paper. Both models perform acceptably on calm days, but only the ANN holds up well on erratic onesand in real WSN deployments, the hardest days for power management are precisely the unpredictable ones where energy shortages are most likely to occur.

V. Discussion

A. Why the ANN Wins on Unpredictable Days

Profile-matching methods make a bet at the start of the day ,they pick a historical pattern that looks similar and stick with it, with only a small correction factor to handle deviations. If the weather changes halfway through the day, that chosen pattern becomes increasingly wrong, and the correction can only do so much. This is a fundamental flaw, not something you can fix by tweaking the settings. The ANN works differently. At every single hour, it looks at what's actually happening right now , including what the irradiance was just one hour ago. So if a cloud suddenly rolls in at 2pm and irradiance drops sharply, the model sees that drop and immediately factors it into the 3pm prediction. It doesn't need to "realize" the day has changed it just reacts to what it directly observes. No amount of fine-tuning a profile-based method can replicate that.

B. Connection to Battery Energy Level

The RMSE gap between the models have effect on network cells. An RMSE of 99 W/m^2 means the Pro-Energy predictor can be wrong by nearly 100 watts per square metre on a typical slot–a significant fraction of the PV panel's full output. A PMU scheduling transmissions on the basis of this prediction will repeatedly over-draw the battery on cloudy afternoons or under-utilise it on unexpectedly clear ones. An RMSE of 20 W/m^2 allows the PMU to operate within a much tighter energy budget, which translates directly into a higher average battery energy level E_L and longer network lifetime [1].

C. Practical Deployment Considerations

The ANN requires a one-time offline training step on historical data. Once trained, inference involves a forward pass through 2,657 parameters(roughly 10,000 floating-point multiply-add operations) which comfortably executes in under a millisecond on any modern embedded processor. Modified Pro-Energy, while simpler to describe, requires maintaining a $D \times N$ memory matrix and running $O(D \times N)$ profile-ranking computations on-device. On very severely constrained microcontrollers, the ANN actually becomes competitive in terms of runtime overhead while delivering far better prediction accuracy.

D. Insights from Short-Term versus Long-Term Prediction

It is important to distinguish between short-term micro-predictions and longer-term macro-predictions when interpreting forecasting performance. Methods operating on very fine time intervals benefit from high temporal autocorrelation in solar irradiance ,conditions typically change only gradually from one minute to the next. In such cases, even simple persistence-based approaches can achieve very low error rates because the next value is often highly similar to the current one.

In contrast, predictions over longer intervals (such as hourly) must account for more substantial changes in weather, including sudden cloud passages or diurnal transitions. For Wireless Sensor Networks, reliable forecasting over extended horizons is usually more valuable, as nodes must plan duty cycles, routing, and transmission decisions for many hours ahead, including overnight periods. Therefore, while short-term predictors may report extremely low errors due to persistence effects, algorithms designed for practical WSN energy management should be evaluated primarily on their ability to handle realistic longer-term variability.

Table V: Statistical Metrics of Signal Volatility ($N = 1000$)

Metric	Minute-Level	Hourly-Level	Ratio (H/M)
Variance of Change	535.07	22848.30	42.70
Volatility (Std Dev)	23.13	151.16	6.53
Mean Abs Change	11.33	121.99	10.77
Max Jump	175.93	720.66	4.10

Unlike previous methods that work well on minute-level data, our work focuses on hourly irradiance prediction, which is much more difficult. Older methods like EWMA, WCMA, and Pro-Energy depend on smooth and stable changes in data, so they perform well when the data changes slowly over time. But in hourly data, the changes are much larger and more random. This is clearly shown in our results. The variance of change increases from 535.07 to 22848.30 (about 42.7 times

higher). The mean absolute change increases from 11.33 to 121.99 (about 10.77 times higher). The standard deviation also becomes 6.53 times larger, and the maximum jump increases from 175.93 to 720.66. This shows that hourly data has much bigger and more sudden changes compared to minute-level data. Because of this, older methods struggle to make good predictions. However, our ANN model can still learn these complex patterns and give better results even when the data is highly unstable.

E. Limitations and Threats to Validity

Several limitations deserve explicit acknowledgement. First, both models were trained and evaluated at a single geographic location (the NREL test site in Colorado). Different climatic profilestropical, arid, temperatemay shift the relative rankings, and the ANN would need retraining for each new deployment site. Second, the ANN requires roughly 4,000 hours of historical data to perform well. Newly installed nodes need a *cold-start* period during which a simpler fallback method should be used. Third, our results are at hourly granularity; performance at finer (15-minute) or coarser (6-hour) resolution remains an open question. Fourth, we evaluate single-step-ahead prediction only.

VI. Conclusion

This paper set out to answer a focused question, does a data-driven ANN outperform the state-of-the-art rule-based Modified Pro-Energy method for solar irradiance prediction in WSNs, and if so, by how much.

By using physics-motivated features like cyclic temporal encodings(day-night, seasonal), and a one-hour lag for previous context and training a small two-hidden-layer network to sidestep overfitting on 4,419 samples, we achieved an R^2 of 0.9867 and an RMSE of 20.027 W/m^2. The Modified Pro-Energy baseline, run under its best-reported hyper parameters, landed at an R^2 of only 0.8203 and an RMSE of 73.721 W/m^2 under the same evaluation conditions.

Moreover, the ANN performs better in conditions like on cloudy, unpredictable days where clouds cause sudden spikes and drops in sunlight that models based historical pattern may struggle to predict accurately, because pattern may be new.

In future work, we plan to use Artificial Neural Network (ANN) to predict solar energy more accurately in wireless sensor networks. ANN can learn from past solar data and give better predictions even when weather is not stable. These better predictions can help the network choose the right cluster head by knowing how much energy each node will harvest in the next round. Also, the scheduling process can use these predictions to decide which nodes should be active and which should sleep. This way, important target areas will always be monitored by nodes that have enough energy. Nodes with low energy can sleep and recharge using solar harvesting. By combining ANN-based prediction with routing and scheduling, we expect the network to last longer, use less energy, and send data to the base station more reliably. This combined approach will also support green IoT network operation.

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